NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 1

CANDIDATE NAME

BIOLOGY CLASS

REGISTRATION NUMBER

BIOLOGY

8875/01

Paper 1 Multiple Choice

19 Sep 2013

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

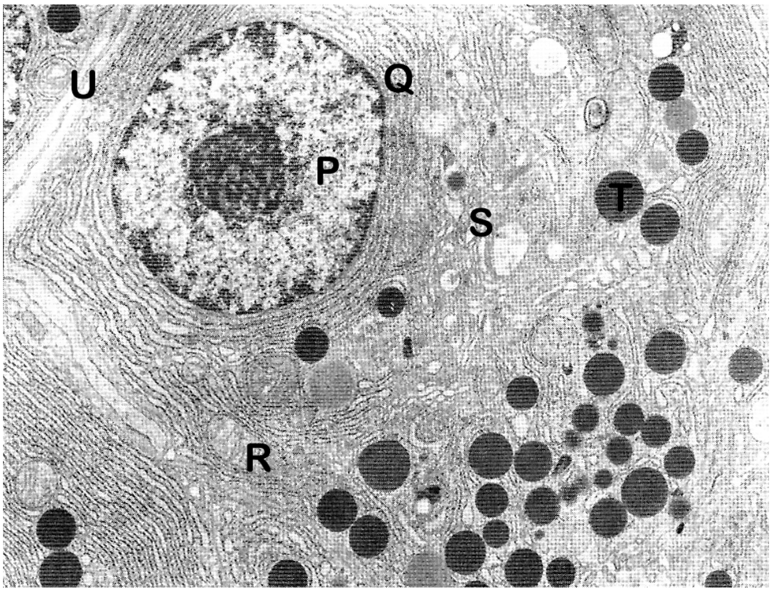
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Biology class, and registration number above and on the Answer Sheet provided.

There are thirty questions on this paper. Answer all questions. For each question there are four possible answers A, B, C and D.
Choose the one you consider correct and record your choice in soft pencil on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

The diagram shows an electron micrograph of a human cell.



Which of the following correctly identifies its structure-function adaptation?

	function	structural adaptation
A	enzyme production	**P and U**
B	antibody production	**Q**, **S** and **T**
C	phagocytosis	**R**
D	photosynthesis	**T**

2 Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it as applied to the roots of the plant. As the first step on finding out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

Which of the following statements are consistent with the scientist's observations?

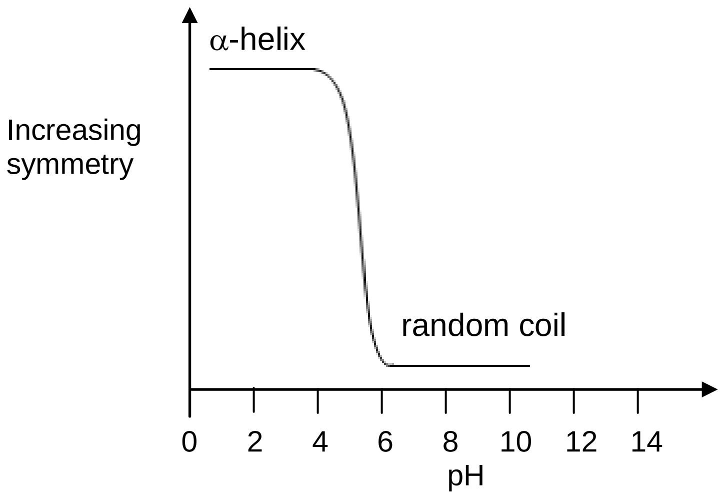
I. The toxin was stored in the central vacuole.
II. The toxin cannot cross the membrane of the organelle in which it is stored.
III. The toxin was stored in chloroplast.
IV. The toxin is likely to be lipid-soluble.
V. The toxin may be an enzyme.

A I, II and V
B I, IV and V
C II, III and IV
D III, IV and V

3 Which row is correct for each of the molecules?

	β-glucose	collagen	haemoglobin	sucrose
A	hexose sugar with a molecular formula $C_6H_{12}O_6$	structural function, found in tendons and blood vessel walls	contains the elements carbon, hydrogen, iron, nitrogen and sulfur	formed by releasing a molecule of water in a hydrolysis reaction
B	repeating monomer of the polysaccharide, cellulose	a molecule consists of three polypeptide chains, each containing a prosthetic group	each non-protein haem group contains a central iron ion	composed of two monosaccharides linked by a glycosidic bond
C	monomer of the 1,6 glycosidic branches of the polysaccharide, glycogen	molecules lie parallel to each other, with cross-links and staggered ends	has two identical α chains and two identical β chains	formed by condensation of two identical monosaccharides
D	in its ring structure, the hydroxyl group of carbon atom 1 is above the plane of the ring	polypeptide chains interact to produce a fibrous protein	has all four levels of protein structure and at least four types of bond	digestion yields glucose and fructose in equal proportions

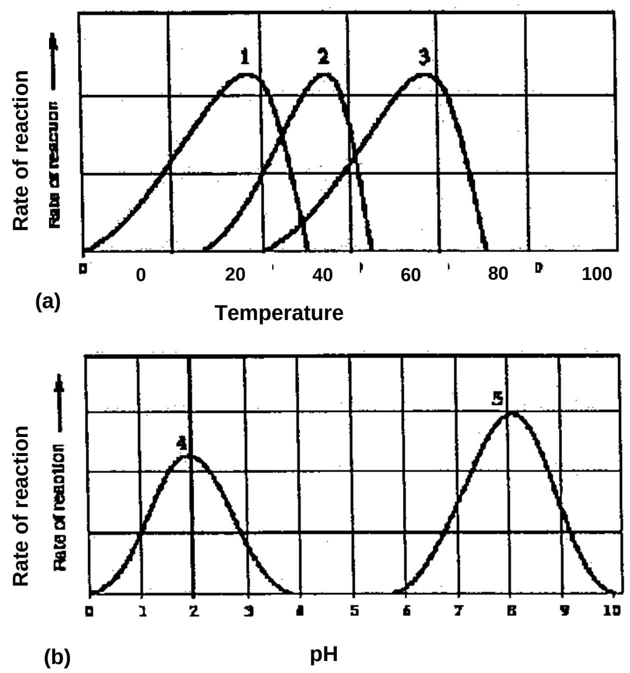
4 The graph shows the effect of pH on the structure of a protein which consists of entirely of repeating residues of one amino acid.



Which statement is true?

A At high acidity the protein loses its secondary structure.
B At high acidity the protein loses its tertiary structure.
C At high acidity the protein loses its tertiary structure.
D At low acidity the protein loses its secondary structure.

5 The following graphs show the activities of different enzymes (1-5) under different conditions:



Which statement is a correct explanation for the rate of reaction of different enzymes?

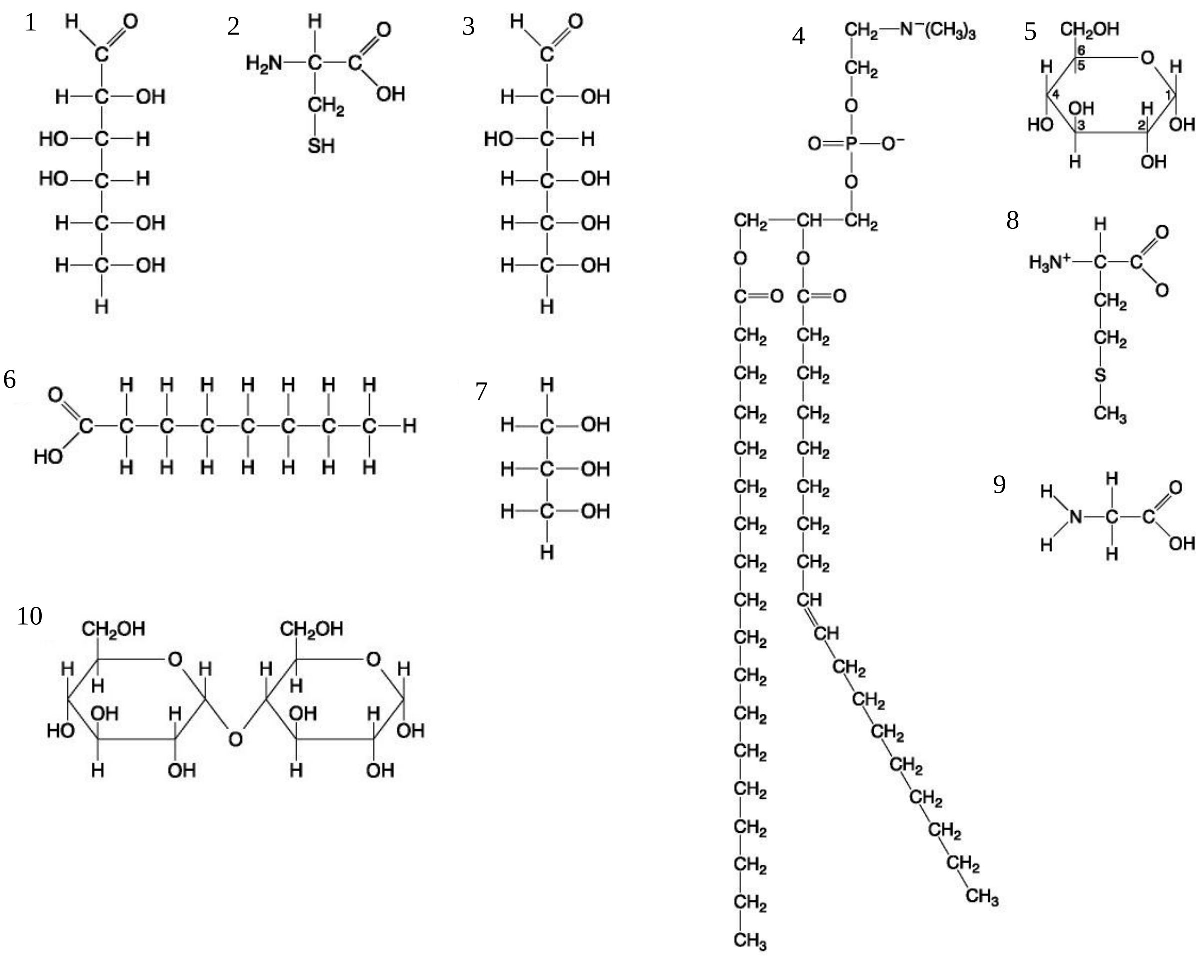
A Rate of reaction of enzyme 3 and its substrate is fastest at 58°C.

B At pH 2, most of the R groups at the active site of enzyme 4 are all negatively charged.

C At pH 8.1, substrate is bonded to the active site of enzyme 5 by hydrogen bonds only.

D At 60°C, several hydrogen bonds between R groups of enzyme 2 are broken.

6 The figure below shows some biomolecules numbered 1 to 10.



Which of the following statements concerning the biomolecules is/are **false**?

I. Triacylglycerol may be made up of one molecule of 6 and three molecules of 7.
II. Molecules 5 and 10 may be joined together by a glycosidic linkage to form a disaccharide.
III. Molecules 1 and 3 are saturated fatty acids while molecule 4 has an unsaturated chain.
IV. All of the molecules are monomers of biological macromolecules.

A	III only
B	I and II only
C	I, II and IV only
D	All of the above

7 The following events occur during the mitotic cell cycle.

I division of centromeres
II proofreading mechanism in replication
III duplication of chromosomes
IV overlapping of non-kinetochore spindle fibres

Which event(s) in mitosis directly ensure that the daughter cells are genetically identical to each other and to the parent cell?

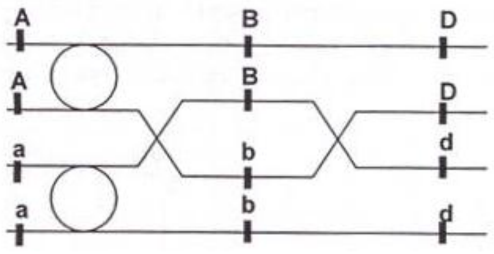
A I only
B I and III
C II and III
D II and IV

8 The diagram shows the chromosomes of one cell which has been squashed during mitosis.

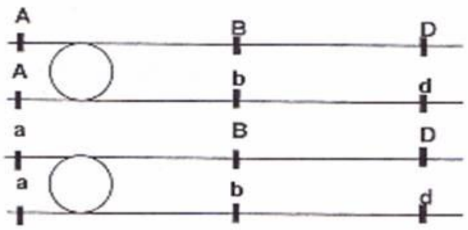
Which stage of mitosis is shown and what is the haploid chromosome number in this species?

	stage of mitosis	haploid chromosome number
A	anaphase	5
B	anaphase	10
C	metaphase	5
D	metaphase	10

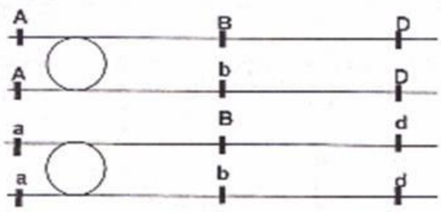
9 The diagram above shows a bivalent. Which of the following correctly represents the final products at the end of Meiosis II?



A



B



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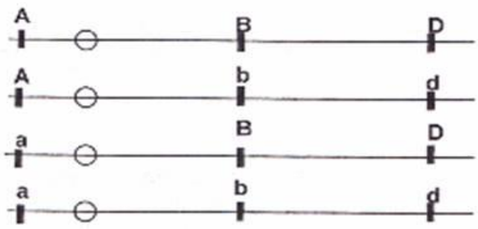
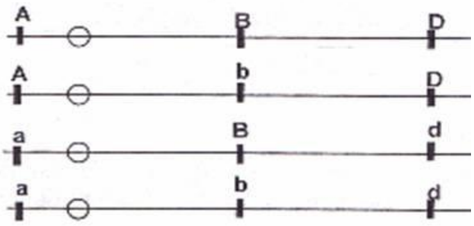


D



10 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- Aphidicholine inhibits DNA polymerase
- Cytarabine is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms
- Epirubicin inhibits an enzyme involved in the unwinding of DNA and separation of strands
- Hydroxycarbamide inhibits an enzyme involved in the production of deoxyribonucleotides

Which row correctly matches a drug to an explanation of the mechanism of action?

	explanation of mechanism of action			
	decreased pool of available nucleotides inhibits chain elongation	DNA strands not available as templates for transcription	DNA damaged during replication and cell death occurs	exposed DNA template strands unable to be copied
A	aphidicholine	epirubicin	cytarabine	hydroxycarbamide
B	epirubicin	cytarabine	hydroxycarbamide	aphidicholine
C	hydroxycarbamide	aphidicholine	epirubicin	cytarabine
D	hydroxycarbamide	epirubicin	cytarabine	aphidicholine

11 A polypeptide molecule contains the amino acid sequence, glycine – leucine – lysine – valine. The table shows the DNA codes for these amino acids.

glycine	leucine	lysine	valine
CCC	GAA	TTT	CAA

Which anticodons present on tRNA are needed for the synthesis of this polypeptide?

A CCC GAA TTT CAA
B CCC GAA UUU CAA
C GGG CUU AAA GUU
D GGG CUU UUU GUU

12 The diagram shows the results of two types of mutation.

C G A T T A C C A T G C A T A
original base sequence

C G A T T A A C C T G C A T A
mutant sequence 1

C G A T T A C C A G G C A T A
mutant sequence 2

What types of gene mutation produce mutant sequences 1 and 2?

	1	2
A	deletion	insertion
B	insertion	deletion
C	inversion	substitution
D	substitution	insertion

13 BRCA1 is expressed in the cells of breast and other tissue, where it helps repair damaged DNA, or destroy cells if DNA cannot be repaired. BRCA1 gene is on chromosome 17 of the human genome.

Which individual is most likely to develop breast cancer?

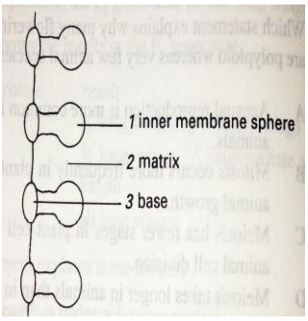
A Has a frame-shift mutation of the BRCA1 allele on each chromosome 17.

B Has a BRCA1 allele on each chromosome 17.

C Has a BRCA1 allele on one chromosome 17.

D Has a BRCA1 allele on one chromosome 17 and another, by translocation, on one chromosome 12.

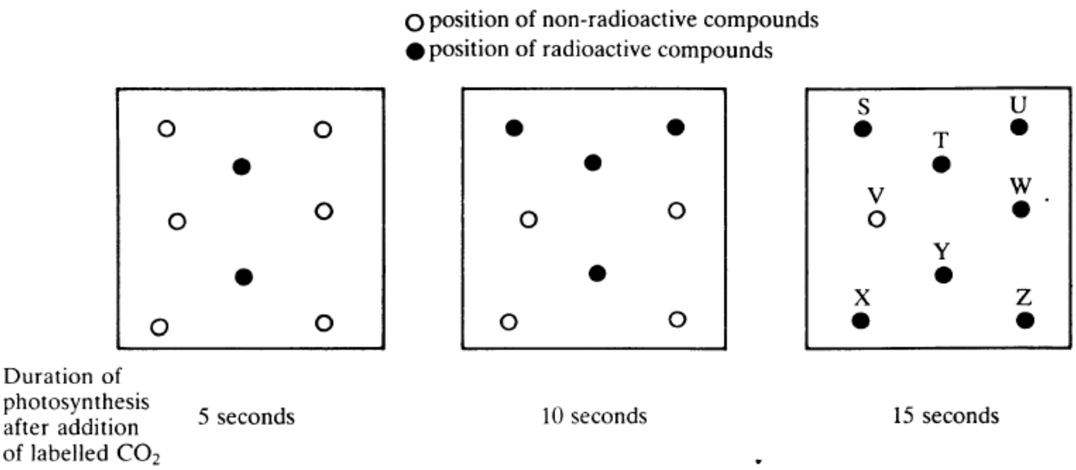
14 The diagram shows stalked particles on part of a crista membrane in a mitochondrion.



What occurs in each of the numbered regions?

	1	2	3
A	ADP synthesis	electron transport	Krebs cycle
B	ATP synthesis	glycolysis	Krebs cycle
C	ATP synthesis	Krebs cycle	electron transport
D	ATP synthesis	Krebs cycle	glycolysis

15 The diagram represents autoradiograph two-dimensional chromatograms of different algal extracts. The extracts were obtained after algal suspensions, which had been exposed to radioactive CO_2 for different length of time in conditions suitable for photosynthesis, were plunged into hot ethanol to prevent further biochemical reactions from occurring.



Assuming that all techniques and conditions (apart from time) were standardised, which of the following conclusions is justified from these data?

A compound X must be sucrose

B compound V must be an intermediate in the photosynthetic pathway

C compound T precedes compound Y in the photosynthetic pathway

D compound U precedes compound X in the photosynthetic pathway

16 Which one of the following summary reactions (each a part of the process of metabolic breakdown of glucose) ultimately yields the most number of ATP molecules, under aerobic conditions? Assume that all electrons complete their trip down the respiratory electron chain.

A 2 pyruvic acid $\longrightarrow$ 2 acetyl-Co-A

B glucose $\longrightarrow$ 2 pyruvic acid

C 2 acetyl-Co-A $\longrightarrow$ 4 carbon dioxide + water

D 2 PGAL $\longrightarrow$ 2 pyruvic acid

17 Cyanide blocks the respiratory electron-transport chain. As a result

A the Krebs cycle speeds up.
B the electrons and hydrogen cannot flow from NAD $_{re}$ to oxygen
C production of water increases
D glycolysis is inhibited.

18 Six tubes containing preparations from animal tissue were set up as shown in the table.

Tube	*Contents*
1	Glucose + homogenized cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvic acid + homogenized cells
5	Pyruvic acid + mitochondria
6	*Pyruvic acid + cytoplasm lacking organelles*

After incubation, in which three tubes would carbon dioxide be produced?

A 1, 2 and 3 **C** 3, 4 and 6
B 1, 4 and 5 **D** 3, 5 and 6

19 The feather colors in budgies (a kind of bird) are determined by two pairs of unlinked genes Y/y and allele B/b. The genotype of green color feather is Y_B_, blue yyB_, and white _ _bb (where _ indicates the presence of either the dominant or recessive allele).

If two heterozygous budgies are mated, what proportion of phenotypes can be expected in the progeny?

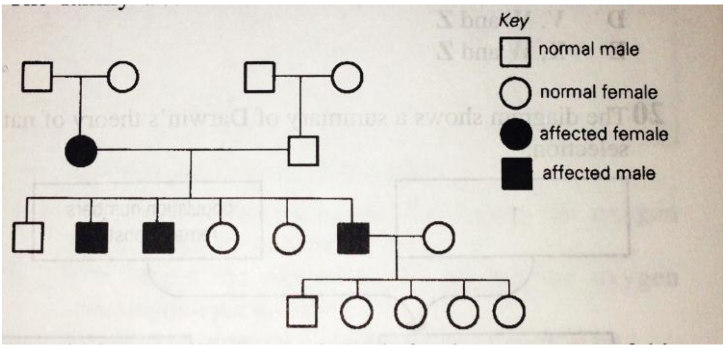
A 9 green color feathers : 7 white color feathers

B 9 green color feathers : 3 blue feathers : 4 white feathers

C 15 green color feathers : 1 blue feathers

D 15 green color feathers : 1 white feathers

20 The family tree shows the inheritance of a human defect.



What is the most likely genetic basis for the occurrence of this defect?

A autosomal dominant
B autosomal recessive
C sex-linked dominant
D sex-linked recessive

21 Red-green colour blindness in mammals is a sex-linked recessive trait. If a colour-blind female is crossed with a normal male, then the theoretical ratio of the offspring will be

A 50% carrier females, 50% colour-blind males.
B 50% normal females, 50% colour-blind males.
C 50% carrier females, 50% normal males.
D 25% carrier females, 25% normal females, 50% normal males.

22 The followings are possible events that may lead to speciation.

I Gene mutation
II Increased gene flow
III Natural selection
IV Geographical isolation
V Habitat differentiation within the same geographical location

The correct order of events that may lead to speciation is

A IV, I, III
B IV, II, III
C V, I, III
D V, II, III

23 Members of two different species possess a similar-looking structure that they use in a similar fashion to perform the same function. Which information would best help distinguish between an explanation based on homology versus one based on convergent evolution?

A Both species are well adapted to their particular environments.

B The sizes of the structures in adult members of both species are similar.

C The two species live at great distance from each other.

D The two species share many proteins in common, and the nucleotide sequences that code for these proteins are almost identical.

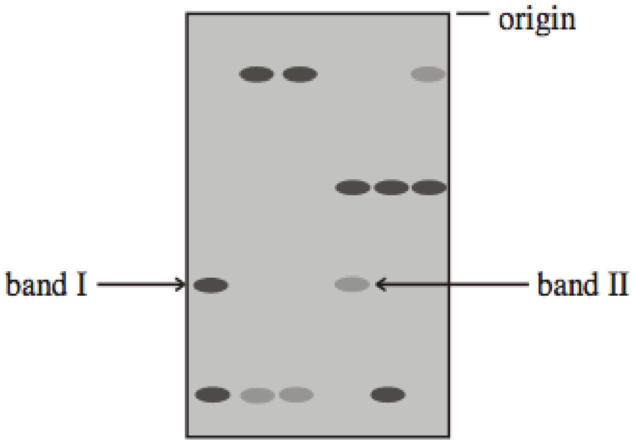
24 Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?

A Comparative embryology.
B Homologous structures.
C Molecular comparisons of DNA and amino acid sequence.
D The fossil records

25 Which feature of stem cells obtained from blood in the umbilical cord enables their use in the treatment of a variety of blood disorders?

A They are totipotent.
B They can differentiate into any type of blood cell.
C They can differentiate into stromal stem cells cells.
D They can renew indefinitely to replace defective blood cells.

26 The diagram below shows the results of DNA profiling using gel electrophoresis.



What conclusion can be drawn about the DNA in bands I and II?

A The DNA in the two bands has the same base sequence.

B The DNA in the two bands consists of fragments of the same length.

C The DNA in the two bands has the same ratio of bases.

D The DNA in the two bands came from the same source.

Refer to the following information to answer **Q27** and **28**.

Erythropoietin (EPO) is a small signaling molecule produced by certain kidney cells, which stimulates the production and differentiation of red blood cells. A scientist is interested to clone the cDNA of EPO receptor (EPO-R) from mouse and express it in suitable human host cells. The first step in library construction is cDNA synthesis.

27 Which of the following components show the correct combinations that are required for synthesizing a collection of cDNAs containing copies of EPO-R cDNA?

	DNA ligase	*DNA polymerase*	*RNA polymerase*	*reverse transcriptase*
A	√	×	√	×
B	√	√	×	×
C	×	×	√	√
D	×	√	×	√

28 The cDNAs synthesized in this mixture were converted into double-stranded DNA molecules supplied with Xho restriction endonuclease specific cohesive ends and ligated into the Xho site of a mammalian expression plasmid.

Which of the following sequence should be present in this plasmid if it is to express inserted cDNAs in human cells?

- **A** A bacterial promoter downstream of the Xho site
- **B** A bacterial promoter upstream of the Xho site.
- **C** A mammalian promoter downstream of the Xho site.
- **D** A mammalian promoter upstream of the Xho site.

29 Which of the following is **not** an example of genetically modified organisms?

- **A** Bt corn plants that are insect-resistant.
- **B** Delayed ripening in tomato using anti-sense technology.
- **C** Golden Rice that produce high levels of beta-carotene.
- **D** Inbred Salmon that grow to adult size quickly.

30 Plants are more readily manipulated by genetic engineering than are animal cells because

- **A** a somatic plant cell can often give rise to a complete plant.
- **B** more vectors are available for transferring recombinant DNA into plant cells.
- **C** plant cells have cellulose cell wall which allows substances to pass through easily.
- **D** plant genes do not contain introns.

--- End of Paper ---